



HybridCount : Multi-Scale transformer with knowledge distillation for object counting

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ABSTRACT

This work introduces a novel architecture that integrates a multi-scale Visual Transformer (ViT) encoder with a graph attention network decoder to model contextual relationships in visual scenes. Our approach achieves real-time, parameter-efficient object counting through an innovative Knowledge Distillation framework that integrates density estimation maps with regression-based counting mechanisms. The distillation process optimizes performance through a three-component loss function: encoder loss, decoder loss, and our proposed Dual-Domain Density-Regression Loss (DD-R Loss). This novel loss formulation simultaneously supervises both spatial density distribution and direct count regression, providing complementary learning signals for robust object quantification. A key contribution is our scale-aware token embedding technique and cross-attention fusion across varying receptive fields within the ViT architecture, enabling precise counting in cluttered visual environments. Experiments are conducted on four crowd-counting datasets, two vehicle counting datasets. Our detailed experimental evaluation shows that the proposed method delivers outcomes comparable to SOTA methods in terms of counting accuracy and density estimate precision. The detailed comparisons presented in our results and discussion sections highlight the significant strengths and advantages of our methodology within the challenging domain of visual object counting. Our framework bridges the gap between the representational power of transformer-based models and graph network architectures. The efficiency of our approach enables real-time performance comparable to other CNN based approaches. This combination delivers a comprehensive solution for object counting tasks that performs effectively even in resource-constrained environments.

1. Introduction

Our design combines the power of advanced transformers with the speed required for real-time object counting. Object counting remains a core computer vision challenge, as it needs multi-scale analysis, spatial localization, size estimation, and accurate count prediction. Vision Transformers (ViTs) [1] have shown strong ability to model long-range relationships, but their computation is often too heavy for practical deployment. In contrast, most traditional CNN models process images at one fixed scale, making it difficult to preserve fine details while understanding the overall scene. This balance is essential for reliable scene interpretation and precise object counting

Many works have studied multi-scale feature extraction [2], but current solutions still have three key limitations. First, features across multiple scales are not efficiently fused in transformer models. Second, knowledge transfer between large and lightweight models remains sub-optimal. Third, graph based scene-relationship modeling, density estimation, and count regression are rarely optimized together. Most prior

studies treat these as separate problems, leading to fragmented methods that miss their natural interactions. As a result, object counting approaches typically adopt either density estimation [3] or direct count regression [4], but seldom unify both to exploit their complementary strengths.

This paper presents a dual-path multi-scale Vision Transformer (ViT) architecture integrated with a Graph Attention Network (GAT) to form a teacher-student network framework. In both the teacher and student networks, **scale-aware embedding** and **cross-attention fusion mechanisms** are employed within the encoder stages. These components enable the networks to effectively capture and integrate information across multiple scales and receptive fields.

The flow of knowledge between the teacher and student networks is structured as follows:

1. **Encoder Distillation:** The first mechanism focuses on transferring the rich, hierarchical representations learned by the teacher network's multi-layer ViT encoders to the lightweight single-layer

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encoders of the student network. This is achieved by aligning the intermediate feature maps at corresponding scales, ensuring that the student network inherits the teacher's ability to extract fine-grained and contextualized features.

2. **Graph Distillation:** To preserve the teacher network's understanding of object relationships within a scene, graph distillation is employed. This mechanism ensures that the student network learns to model the relational structure of objects in the scene, enabling it to replicate the teacher's ability to interpret how different elements interact with one another.
3. **Density and Count Distillation:** The final mechanism preserves the spatial zoning capabilities of the teacher network. Through this process, the student network learns to accurately represent the physical distances and relationships between objects, ensuring that it maintains an effective understanding of the spatial layout of scenes.

Together, these three mechanisms create a comprehensive bridge for transferring the intricate knowledge of the teacher network to the more compact student network. By combining scale-aware embedding, cross-attention fusion, and the aforementioned distillation techniques, the proposed architecture achieves a balance between computational efficiency and high performance.

Our primary contributions are:

- A novel multi-scale ViT architecture that effectively combines information across varying receptive fields through scale-aware token embedding and cross-attention fusion mechanisms.
- A sophisticated knowledge distillation architecture facilitates the streamlined transfer of visual comprehension capabilities from resource-intensive models to deployment-optimized alternatives.
- Dual-domain density-regression Loss (DD-R Loss) combines two powerful approaches - density estimation and direct count regression - in a unified optimization framework.
- Extensive experimental validation demonstrating state-of-the-art performance across multiple benchmarks, with significant improvements in efficiency-accuracy trade-offs compared to existing methods.

Our work provides important insights into the design of efficient yet powerful vision systems that combine the strengths of transformer architectures with practical deployment considerations. The techniques we have developed can help solve real-world problems in many areas. They are helpful whenever you need to understand complex scenes or count objects accurately. For example, stores can use them to track inventory, environmental scientists can monitor wildlife populations, and self-driving vehicles can better understand their surroundings. These methods offer practical solutions across diverse fields by bridging the gap between detailed scene analysis and precise light weight counting.

2. Background

Object counting in high-density scenarios has evolved significantly with deep learning methodologies, transitioning from traditional CNN-based approaches to sophisticated transformer architectures. Zhang et al. [5] introduced the foundational Multi-Column CNN (MCNN) that revolutionized crowd counting through parallel columns with different receptive field sizes, establishing multi-scale feature extraction as crucial for accurate density estimation. Li et al. [6] advanced this with CSRNet, utilizing dilated convolutions to maintain spatial resolution while expanding receptive fields, achieving remarkable improvements on dense datasets. Cao et al. [7] further enhanced the paradigm with Scale Aggregation Network (SANet), introducing explicit scale-aware feature extraction through attention mechanisms. The introduction of Vision Transformers by Dosovitskiy et al. [1] revolutionized the field by enabling self-attention mechanisms to capture long-range dependencies without locality constraints. Subsequent works like LoViT-Crowd [8], CCTwins [9], and CrowdCCT [10] demonstrated that combining CNN

features with transformer-based global modeling could significantly improve performance, though computational overhead remained a challenge for real-time deployment.

The pursuit of computational efficiency has driven innovative lightweight architectures that balance performance with resource constraints. Wang et al. [11] developed MobileCount with MobileNetV2 backbone, achieving significant FLOP reductions while maintaining accuracy. Liu et al. [12] proposed Lw-Count with refined ghost blocks and scale regression modules, demonstrating that efficiency could be achieved through careful feature propagation design. Li et al. [13] introduced a light multi-scale feature fusion network that balanced computational efficiency with representation capacity. Simultaneously, knowledge distillation emerged as a powerful paradigm for efficient model creation. Gou et al. [14] established theoretical foundations for teacher-student frameworks, while recent advances by Li et al. [15] and Cho et al. [16] introduced dual teachers and ambiguity-aware robust teacher frameworks for self-knowledge distillation. Specialized applications have shown versatility across domains, with Fu et al. [17] developing LD-Net for detection networks and Ruiping Liu et al. [18] proposing TransKD for structured knowledge transfer. Graph Neural Networks have also gained attention, with C.Zhang et al. [19] developing Crowd-Graph for spatial relationship modeling and X.Guo et al. [20] introducing group and graph attention networks for enhanced counting performance.

2.1. Attention mechanisms

Attention mechanisms enhance vision tasks: Multi-attention network for compressed video referring object segmentation [21], memory-enhanced attention for contrastive learning [22], Rethinking attentive object detection via neural attention learning [23], and structural attention for image synthesis [24]. These methods apply single-purpose attention within their domains.

Our dual-attention design addresses counting-specific challenges: (1) **cross-scale attention** fuses ViT patches at multiple resolutions (32×32 , 16×16 , 8×8) to handle extreme size variation, and (2) **graph attention** operates on spatial topology with learned edges to model distribution patterns.

Despite significant progress, current approaches face fundamental limitations where lightweight methods often sacrifice semantic richness for computational efficiency, limiting effectiveness in complex scenes with irregular distributions and occlusions. Most knowledge distillation methods rely on simple feature mimicking without leveraging structural insights about spatial relationships and density patterns. The integration of different architectural paradigms-CNNs for local features, Transformers for global context, and efficient design principles-remains underexplored, with approaches typically focusing on individual aspects rather than unified frameworks. Our proposed HybridCount addresses these limitations by integrating hybrid architectural design with sophisticated knowledge distillation strategies, creating a lightweight framework that maintains both local feature precision and global contextual understanding while ensuring practical deployability in resource-constrained environments. Recent benchmarks like NWPU-MOC [25] provide large-scale diversity but serve primarily as evaluation datasets rather than model-centric contributions. Domain-adaptive (DAOT [26]), class-incremental (Evolving Object Counting [27]), and class-agnostic methods (DEN [28]) address different problem settings (cross-domain or class generalization), making direct comparison infeasible.

3. Methodology

3.1. Architecture overview

Our methodology presents a scale-aware object counting framework leveraging knowledge distillation within a dual-pathway architecture.

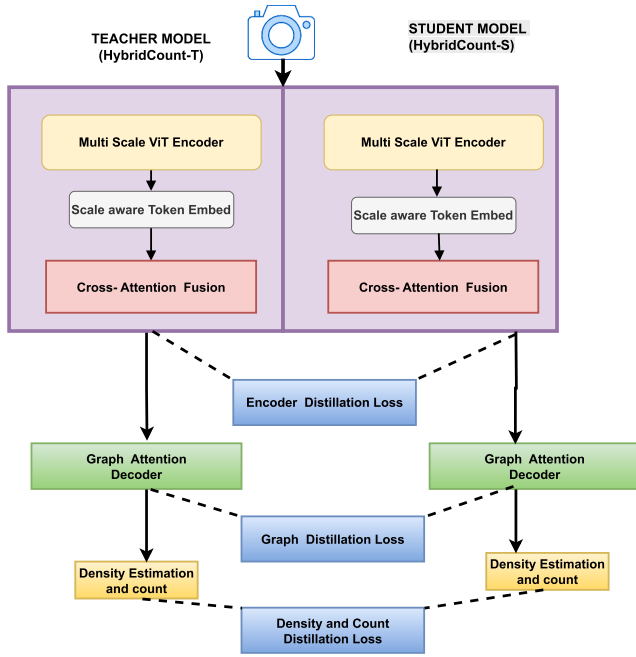


Fig. 1. Overall multi-scale architecture.

The teacher model employs a Multi-Scale Vision Transformer (ViT) Encoder comprising three parallel Scale ViT Encoders (each with 2 layers), processing input images at different scales using varied patch sizes to capture multi-scale features. These multi-scale representations are combined through a Scale-aware Token Embed layer followed by Cross-Attention Fusion to produce a unified feature representation. The Multi-Scale Graph Decoder receives the fused features and constructs a graph that captures spatial relationships between objects; then, a Graph Attention Decoder refines these representations by highlighting relevant spatial patterns. A Density Estimation and Count module processes the refined spatial features to predict both a density map and a total count value. This entire architecture is replicated in the student model with a lightweight implementation (only 1 layer per Scale ViT Encoder), maintaining structural consistency while significantly decreasing parameter count. Knowledge transfer occurs through three strategically placed distillation losses: Encoder Distillation Loss aligning the fused ViT representations, Graph Distillation Loss connecting the refined spatial features from Graph Attention Decoders, and Depth Distillation Loss ensuring consistency between both density maps and count predictions. This comprehensive distillation approach enables the lightweight student model to achieve comparable performance to the parameter-intensive teacher model while maintaining computational efficiency, as visualized in Fig. 1.

3.2. Multi scale visual transformer

Our approach utilizes multi-scale Vision Transformer (ViT) encoders to effectively handle the scale variation challenge inherent in object counting tasks. The multi-scale design enables the model to capture both fine-grained details and global contextual information simultaneously. The teacher model (HybridCount-T) employs a multi-scale ViT encoder with 2 transformer layers at each scale. We process the input image at three different scales using varying patch sizes: smaller patches (e.g., 8×8) for Scale 1 to capture fine details, medium patches (e.g., 16×16) for Scale 2, and larger patches (e.g., 32×32) for Scale 3 to capture global context. Each scale-specific encoder independently extracts features relevant to its respective receptive field, with each transformer layer consisting of multi-head self-attention and feed-forward networks. The student model (HybridCount-S) maintains the same multi-scale structure

Table 1

Parameter and FLOP Comparison of Teacher and Student ViT Encoders. FLOPs assume 196, 49, and 16 tokens for Scales 1–3.

Component	T-Params	S-Params	Red.%	T-FLOPs	S-FLOPs	Red.%
<i>Scale 1 Per Layer (196 tokens)</i>						
Q, K, V proj.	786k	12k	98.40	308M	4.80M	98.40
Output proj.	262k	4k	98.40	103M	1.60M	98.40
Attention	–	–	–	39M	2.50M	93.80
FFN 1st layer	1,049k	8k	99.20	412M	6.40M	98.40
FFN 2nd layer	1,049k	8k	99.20	412M	6.40M	98.40
LayerNorms	2k	0.3k	87.50	1.20M	0.20M	87.50
Patch embed	98k	12k	87.50	39M	4.80M	87.50
Per layer	3,246k	45k	98.60	1,313M	27M	98.00
Scale 1	6,492k	45k	99.30	2,626M	27M	99.00
Scale 2	6,492k	35k	99.50	657M	6.70M	99.00
Scale 3	6,492k	30k	99.50	214M	2.20M	99.00
Token Embed	500k	15k	97.00	98M	2.90M	97.00
Cross-Attn Fusion	1,200k	25k	97.90	235M	4.90M	97.90
Total Encoder	21,176k	150k	99.30	3,830M	43M	98.90

Note: T-Params = Teacher Parameters, S-Params = Student Parameters.

Teacher: $d_{\text{model}} = 512$, $d_{\text{ff}} = 2048$; Student: $d_{\text{model}} = 64$, $d_{\text{ff}} = 256$.

but with a significantly reduced parameter count, implementing only 1 transformer layer at each scale. This lightweight configuration preserves the model's ability to process multiple scales while substantially improving computational efficiency. To integrate information across scales, we implement a scale-aware token embedding mechanism that projects tokens from different scales into a unified embedding space while preserving scale-specific information through learnable scale encodings. These embedded tokens from all scales then pass through a cross-attention fusion module, where tokens from different scales attend to each other, allowing for information exchange between fine and coarse representations. This fusion mechanism employs multi-head cross-attention where queries from one scale attend to keys and values from all scales, creating a comprehensive representation that maintains semantic similarity of features across all scales. The computational advantage of our student model becomes evident in this multi-scale approach, requiring approximately half the parameters of the teacher model while maintaining the essential scale-aware processing capabilities critical for accurate object counting.

As illustrated in Table 1, our student model achieves a substantial parameter reduction compared to the teacher model while maintaining the essential multi-scale architecture. The detailed parameter breakdown highlights how our lightweight design decisions halving the transformer layers and significantly reducing embedding dimensions result in a 99.3% reduction in total ViT encoder parameters. Fig. 2 provides a visualization of the scale-aware embedding and cross-attention fusion mechanisms that are central to our approach. As shown in the diagram, each scale's features undergo linear projection followed by the addition of scale-specific encodings, ensuring that tokens maintain their scale identity while enabling cross-scale interactions through the multi-head attention mechanism.

3.3. Graph construction and attention mechanism

After extracting multi-scale fused representations from the ViT encoders, we reformulate these features into structured graphs to explicitly capture spatial dependencies among object regions. The graph attention aggregator transforms ViT patch features into spatially-aware embeddings through four sequential stages: (1) hybrid adjacency construction, (2) graph convolution, (3) local graph attention, and (4) global multi-head attention. This hierarchical design captures both local spatial coherence and long-range dependencies essential for density estimation.

3.3.1. Four-stage processing pipeline

- Stage 1: Hybrid Adjacency Matrix: We need to define which patches should communicate with each other in the graph. Combining fixed

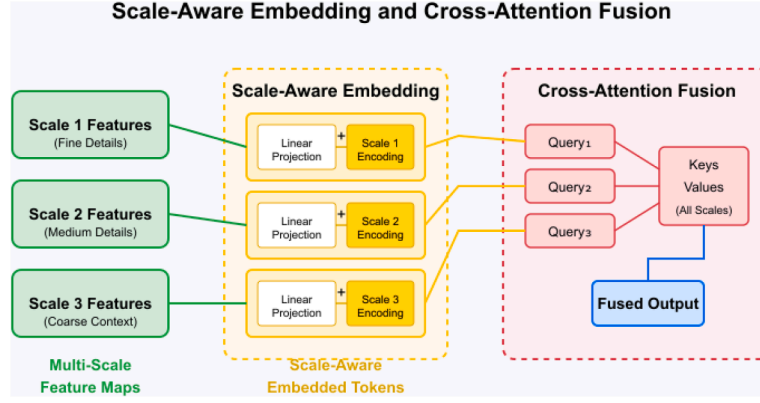


Fig. 2. Scale aware embedding and cross attention.

spatial connections (nearby patches) with learned semantic connections (similar patches anywhere) allows the model to leverage both geometric structure and adaptive feature relationships.

Given fused features $F \in \mathbb{R}^{H \times W \times D}$, we construct graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with $N = H \times W$ nodes and hybrid adjacency:

$$A = \text{Softmax}(A_{\text{base}} + A_{\text{learnable}}), \quad (1)$$

where A_{base} encodes spatial proximity (4- or 8-connectivity) and $A_{\text{learnable}} \in \mathbb{R}^{N \times N}$ learns semantic connections.

- Stage 2: Graph Convolution : Each patch should aggregate information from its connected neighbors to build richer representations. By propagating features along graph edges over multiple layers, nearby regions share contextual information while respecting the spatial-semantic structure defined by the adjacency matrix.

Multi-layer message passing propagates local information:

$$N^{(l+1)} = \sigma(AN^{(l)}W_g^{(l)}), \quad l = 0, \dots, L-1. \quad (2)$$

- Stage 3-Graph Attention (GAT): Recognizing that neighbors differ in informational value, graph attention estimates attention coefficients that prioritize salient spatial relationships and down-weight irrelevant connections.

Adaptive neighborhood aggregation with learned attention:

$$\mathbf{h}'_i = \sigma \left(\sum_{j \in \mathcal{N}(i)} \alpha_{ij} \mathbf{W}_j \mathbf{h}_j \right), \quad \alpha_{ij} = \frac{\exp(\mathbf{a}^T [\mathbf{W}_i \mathbf{h}_i \| \mathbf{W}_j \mathbf{h}_j])}{\sum_{k \in \mathcal{N}(i)} \exp(\mathbf{a}^T [\mathbf{W}_i \mathbf{h}_i \| \mathbf{W}_k \mathbf{h}_k])}. \quad (3)$$

- Stage 4: Global Multi-head Attention: Graph convolution and GAT are limited to local neighborhoods, potentially missing long-range patterns like similar crowd densities in distant image regions. Global attention allows every patch to attend to all others without topological constraints, capturing scene-wide context essential for handling occlusion and perspective distortion.

Unrestricted long-range modeling:

$$\text{Attn}(Q, K, V) = \text{Softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V, \quad Q = NW_Q, K = NW_K, V = NW_V. \quad (4)$$

The complete pipeline is detailed in Algorithm 1, with parameter analysis in Table 2.

Configuration.. Teacher: $D=320 \rightarrow d=256$, $L=3$, $H=10$ heads. Student: $D=80 \rightarrow d=64$, $L=3$, $H=4$ heads. This achieves 93% parameter reduction while preserving spatial reasoning capability.

3.4. Dual-domain density-regression (DD-R) loss

The proposed Dual-Domain Density-Regression (DD-R) loss jointly supervises the model in both spatial and global domains, allowing

Algorithm 1 Four-stage graph attention decoder.

Require: $F \in \mathbb{R}^{H \times W \times D}$ (ViT features)

Ensure: $G_{\text{out}} \in \mathbb{R}^{N \times d}$ (graph embeddings)

- 1: **Stage 1: Graph Construction**
- 2: $N^{(0)} \leftarrow \text{Linear}(F) \in \mathbb{R}^{N \times d}$, $N=H \times W$ $\mathcal{O}(N D d)$
- 3: $A_{\text{base}} \leftarrow \text{Spatial4Connect}(H, W)$ $\mathcal{O}(N)$
- 4: $A \leftarrow \text{Softmax}(A_{\text{base}} + A_{\text{learnable}})$ $\mathcal{O}(N^2)$
- 5: **Stage 2: Graph Convolution** ($L=3$ layers)
- 6: **for** $l = 0$ to $L-1$ **do**
- 7: $N^{(l+1)} \leftarrow \text{ReLU}(AN^{(l)}W_g^{(l)})$ $\mathcal{O}(N^2 d)$
- 8: **end for**
- 9: **Stage 3: Graph Attention (GAT)**
- 10: **for** $i = 1$ to N **do**
- 11: $\alpha_i \leftarrow \text{SoftmaxAttn}(\{N_j^{(L)} : j \in \mathcal{N}(i)\})$ $\mathcal{O}(Ed)$
- 12: $N_i^{\text{GAT}} \leftarrow \sigma(\sum_{j \in \mathcal{N}(i)} \alpha_{ij} W N_j^{(L)})$
- 13: **end for**
- 14: **Stage 4: Global Multi-head Attention** ($H=10$ heads for teacher, 4 for student)
- 15: $Q, K, V \leftarrow N^{\text{GAT}} W_Q, N^{\text{GAT}} W_K, N^{\text{GAT}} W_V$ $\mathcal{O}(N d^2)$
- 16: $G_{\text{out}} \leftarrow \text{MultiHead}(Q, K, V) W_O \in \mathbb{R}^{N \times d}$ $\mathcal{O}(N^2 d)$
- 17: **return** G_{out}

Table 2

Graph network configuration: parameters, dimensions, and computational cost per stage.

Stage	Component	Dim	Teacher	Student	Red.%	Cost
1	Proj: $F \rightarrow N^{(0)}$	$N \times d$	81,920	5120	93.8	$\mathcal{O}(N D d)$
	A_{base}	$N \times N$	0	0	–	$\mathcal{O}(N)$
	A_{learn}	$N \times N$	65,536	4096	93.8	$\mathcal{O}(N^2)$
2	GCN ($L=3$)	$N \times d$	196,608	12,288	93.8	$\mathcal{O}(L N^2 d)$
3	GAT weights W	$d \times d$	65,536	4096	93.8	$\mathcal{O}(Ed)$
	GAT attn a	$2d$	512	128	75.0	
4	Q, K, V proj	$3d \times d$	245,760	15,360	93.8	$\mathcal{O}(N d^2)$
	Heads: $H \times d_k$	–	10×32	4×16	87.5	$\mathcal{O}(N^2 d)$
	Output W_O	$d \times d$	81,920	10,240	87.5	$\mathcal{O}(N d^2)$
Total Parameters			737,792	51,328	93.0	
Output: G_{out}			$N \times d$	256×256	256×64	75.0

the network to learn accurate local density distributions while maintaining global counting consistency. Given a training dataset $\mathcal{D} = \{(I_i, D_i^{\text{gt}}, C_i^{\text{gt}})\}_{i=1}^N$, each image I_i has a corresponding ground-truth density map $D_i^{\text{gt}} \in \mathbb{R}^{H \times W}$ and total object count $C_i^{\text{gt}} = \sum_{x,y} D_i^{\text{gt}}(x, y)$.

The network outputs both a predicted density map D_i and a directly regressed count C_i^{pred} , obtained from the respective density and regression heads. The integrated count from the predicted density is computed

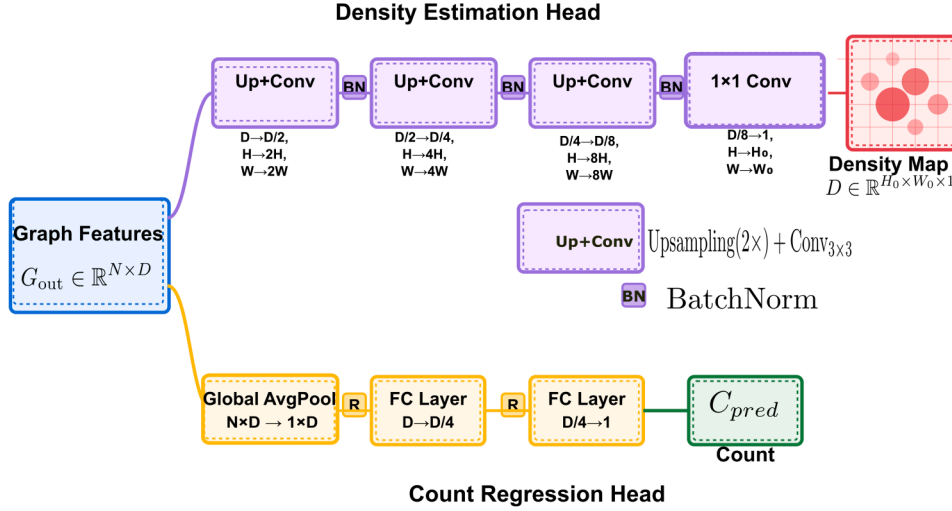


Fig. 3. Overview of the proposed dual-headed density-count estimation framework. A Graph Attention Network generates fused token features $G_{out} \in \mathbb{R}^{N \times D}$, shared by both heads. The Density Head (violet) applies three stages of Upsample (2 $\times$) $\rightarrow$ 3 $\times$ 3 Conv $\rightarrow$ BatchNorm with channel reduction ($D \rightarrow D/2 \rightarrow D/4 \rightarrow D/8 \rightarrow 1$) to predict the full-resolution density map $D \in \mathbb{R}^{H_0 \times W_0 \times 1}$, where the color bar indicates spatial density per cell. In parallel, the Count Head (yellow) performs Global Average Pooling over N followed by FC layers ($D \rightarrow D/4 \rightarrow 1$) to regress the scalar count $C_{pred} \in \mathbb{R}$, enabling spatial and global counting objectives in one forward pass. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

as:

$$C_i^{dens} = \sum_{x,y} D_i(x, y). \quad (5)$$

The DD-R loss comprises three complementary components:

(1) *Density-domain loss* $\therefore$ A pixel-wise mean squared error (MSE) enforces spatial accuracy between the predicted and ground-truth density maps:

$$\mathcal{L}_{dens} = \frac{1}{N} \sum_{i=1}^N \frac{1}{HW} \|D_i - D_i^{gt}\|_2^2. \quad (6)$$

(2) *Count-domain loss* $\therefore$ A regression loss enforces global numerical accuracy between the predicted and ground-truth total counts:

$$\mathcal{L}_{count} = \frac{1}{N} \sum_{i=1}^N |C_i^{pred} - C_i^{gt}|. \quad (7)$$

(3) *Cross-domain consistency loss* $\therefore$ To ensure semantic agreement between the spatial (density-based) and global (regression-based) predictions, we impose a soft regularization term that encourages the integrated density count to align with the directly regressed count:

$$\mathcal{L}_{cons} = \frac{1}{N} \sum_{i=1}^N (C_i^{dens} - C_i^{pred})^2. \quad (8)$$

The overall DD-R loss is then expressed as a weighted combination:

$$\mathcal{L}_{DD-R} = \omega_{dd} \mathcal{L}_{dens} + \omega_{cd} \mathcal{L}_{count} + \omega_{cdc} \mathcal{L}_{cons}, \quad (9)$$

where ω_{dd} , ω_{cd} , and ω_{cdc} are balancing coefficients that control the contributions from the density, regression, and consistency domains, respectively. In our experiments, we set $\omega_{dd} = 0.7$, $\omega_{cd} = 0.25$, and $\omega_{cdc} = 0.05$ based on validation performance.

The DD-R loss effectively bridges the distributional and scalar supervision signals. The density-domain term promotes spatial precision, the count-domain term enhances global accuracy, and the cross-domain term maintains consistency between them. This dual supervision stabilizes convergence and improves the robustness of the model across varying object densities and scales. The structure of DD-R Loss calculation network shown in the Fig. 3.

3.5. Knowledge distillation framework

Our knowledge distillation framework enables the lightweight student model to learn from the more powerful teacher model while maintaining significantly reduced parameters. We implement a comprehensive three-level distillation approach, targeting different stages of the network to ensure effective knowledge transfer.

The distillation process begins after the teacher model has been fully trained on the object counting task. During distillation, we freeze the teacher model's parameters and train only the student model using a combination of task-specific losses and distillation losses.

For the encoder distillation, we align the multi-scale feature representations after the cross-attention fusion module. Given the teacher's fused features F_T and the student's fused features F_S , we define the encoder distillation loss as:

$$L_{enc} = \frac{1}{N} \sum_{i=1}^N \left\| \frac{F_T^i}{\|F_T^i\|_2} - \frac{F_S^i}{\|F_S^i\|_2} \right\|_2^2 \quad (10)$$

where N is the number of feature vectors and the normalization ensures that the loss focuses on directional similarity rather than magnitude.

For the graph distillation, we transfer knowledge from the teacher's refined graph features to the student's. Given the teacher's graph attention output G_T and the student's graph attention output G_S , the graph distillation loss is:

$$L_{graph} = \frac{1}{M} \sum_{i=1}^M \|G_T^i - G_S^i\|_2^2 \quad (11)$$

where M is the number of nodes in the graph representation.

For the depth distillation, we align both the density maps and count predictions. Given the teacher's density map D_T and count C_T , and the student's density map D_S and count C_S , we define:

$$L_{den_distill} = \|D_T - D_S\|_2^2 \quad (12)$$

$$L_{cnt_distill} = C_T - C_S \quad (13)$$

Additionally, we incorporate direct supervision from ground truth to ensure the student model learns both from the teacher and from real data:

$$L_{den_gt} = \|D_{gt} - D_S\|_2^2 \quad (14)$$

$$L_{cnt_gt} = \|C_{gt} - C_S\| \quad (15)$$

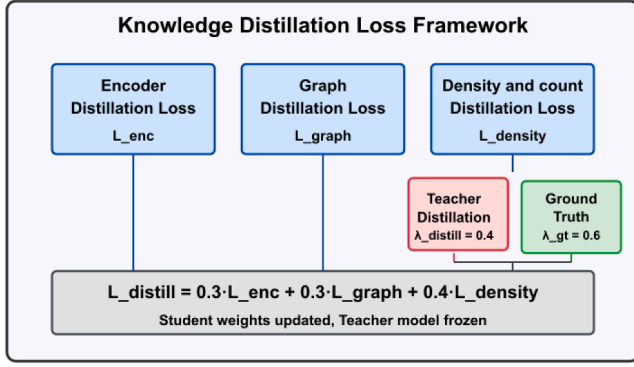


Fig. 4. Distillation loss computation framework.

The combined depth loss is then:

$$L_{\text{depth}} = \lambda_{\text{distill}} \cdot (L_{\text{den_distill}} + L_{\text{cnt_distill}}) + \lambda_{\text{gt}} \cdot (L_{\text{den_gt}} + L_{\text{cnt_gt}}) \quad (16)$$

where λ_{distill} and λ_{gt} are weighting parameters set to 0.4 and 0.6, respectively, giving higher importance to ground truth supervision.

The total distillation loss is a weighted combination of these three components:

$$L_{\text{distill}} = \alpha_{kd} \cdot L_{\text{enc}} + \beta_{kd} \cdot L_{\text{graph}} + \gamma_{kd} \cdot L_{\text{depth}} \quad (17)$$

where α_{kd} , β_{kd} , and γ_{kd} are set to 0.3, 0.3, and 0.4, respectively, based on validation performance.

This comprehensive distillation framework ensures that the student model learns both from ground truth supervision and from the teacher model's representations at multiple levels. The student model's weights are the only ones updated during this process, as the teacher model remains frozen. The result is a highly efficient model that achieves 99.1% parameter reduction while maintaining comparable accuracy to the much larger teacher model. The detailed loss diagram shown in Fig. 4.

4. Experiments

We evaluate the proposed scale-aware distillation framework under standard datasets and protocols.

4.1. Datasets

ShanghaiTech [29]: 1,198 images split into Part A (high density) and Part B (low density); a common benchmark across density regimes.

UCF-QNRF [30]: 1,535 high-resolution images with extremely dense crowds; stresses scalability at high counts.

JHU-CROWD++ [31]: 4,250 images with diverse densities, heavy occlusions, and varied scenes; tests robustness in realistic conditions.

CARPK [32]: 1,448 aerial images of parking lots with 89,777 annotated vehicles; evaluates counting performance on overhead vehicle detection.

PUCPR+ [32]: 125 parking lot images with controlled viewpoints and moderate vehicle density; assesses generalization to structured environments.

4.2. Evaluation metrics

We evaluate our scale-aware distillation framework using standard object counting metrics. The **Mean Absolute Error (MAE)** measures the average absolute difference between predicted and ground-truth counts:

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N \|y_i - \hat{y}_i\|, \quad (18)$$

Table 3

Training hyperparameters for student-teacher knowledge distillation.

Parameter	Value	Parameter	Value
<i>Teacher Model Training</i>			
Teacher epochs	500	Teacher batch size	16
Teacher optimizer	Adam	Teacher LR	10^{-4}
Teacher dropout	0.3	Loss function	MSE
<i>Student Model Training</i>			
Student epochs	300	Student batch size	8
Student optimizer	Adam	Student LR	10^{-4}
Student dropout	0.2	Distillation temp.	3.0
<i>Common Settings</i>			
Weight decay	10^{-5}	Gradient clipping	5.0
LR scheduler	CosineAnnealing	Warmup epochs	10
Validation split	0.15	Early stopping	80
Random seed	42	Mixed precision	FP16

where N is the number of test images, y_i the ground-truth count, and $\hat{y}_i$ the predicted count.

The **Root Mean Square Error (RMSE)** is defined as:

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2}, \quad (19)$$

placing higher emphasis on large deviations and capturing robustness against dense or highly variable scenes.

We also report model parameters (in millions) and inference speed (FPS) to highlight efficiency gains of the student model.

4.3. Implementation details

All experiments are conducted in PyTorch 1.8.0 with CUDA 11.1 on a single NVIDIA V100 GPU. For consistency, identical hardware and hyperparameters are used across all variants.

Training Protocol. The teacher (24M parameters) and student (0.22M) networks are trained sequentially. The teacher is trained for 500 epochs using Adam ($\beta_{\text{grad,smoothing}}=0.9$, $\beta_{\text{adaptive,caling}}=0.999$) with an initial learning rate of 1×10^{-4} and cosine annealing decay. The student is trained for 300 epochs with the same settings, using frozen teacher weights. Distillation losses are weighted as $(\alpha_{kd}, \beta_{kd}, \gamma_{kd})=(0.3, 0.3, 0.4)$ for encoder-, graph-, and depth-level distillation, respectively. The hyperparameters used for training are shown in the Table 3.

Data Preprocessing and Augmentation

Ground-truth density maps are generated using a geometry-adaptive Gaussian kernel following Zhang et al. [5]. For annotated points $\{p_i\}_{i=1}^M$, the density map $D(x)$ is:

$$D(x) = \sum_{i=1}^M \mathcal{N}(x - p_i, \sigma_i^2), \quad (20)$$

where σ_i is the average distance to the $k = 3$ nearest neighbors:

$$\sigma_i = \frac{1}{k} \sum_{j=1}^k \|p_i - p_j\|_2. \quad (21)$$

This produces broader kernels in sparse regions and sharper ones in dense areas, ensuring scale consistency. Images are resized to 224×224 , normalized to $[0, 1]$, and augmented with random flips, crops, and brightness/contrast jittering. Each density map is normalized so that $\sum_x D(x) = C^{\text{gt}}$, preserving the true object count.

Inference Procedure

During inference, full-resolution images are processed directly whenever computational resources allow. For larger inputs, the image is divided into manageable regions that are processed independently, and

their outputs are seamlessly combined to ensure consistency across the full image. This is a standard implementation step for efficient memory usage and does not represent an additional algorithmic component of the model.

Cross-Platform Efficiency

The proposed student model scales effectively across devices, maintaining a favorable balance of speed, memory, and energy use. It achieves **50 FPS** on a 2-core CPU, **92 FPS** on an RTX 3090 GPU, and **27 FPS** on a Jetson Xavier NX. Memory usage is limited to **0.36 GB** on GPU and **180 MB** on CPU, with average power consumption of **65 W** and **12 W** on GPU and edge devices, respectively. This demonstrates practical deployability for both embedded and high-performance platforms. The detailed values are given in Table 4.

5. Ablation studies

We conduct extensive ablation experiments on the ShanghaiTech Part A dataset to validate the contribution of each component in the proposed scale-aware knowledge distillation framework. The studies cover architectural design, distillation strategy, hyperparameter sensitivity, and model size analysis.

5.1. Architecture component analysis

Table 5 compares model variants with different components removed. The complete student model (**HybridCount**) achieves the best performance with **MAE = 52.25** and **RMSE = 85.64**, using only **0.22M parameters**. Removing key modules leads to consistent degradation. The absence of *scale-aware embedding* severely increases error (MAE 71.8), showing its role in multi-scale preservation. Excluding *cross-attention fusion* (MAE 68.7) weakens global-local integration, while removing the *graph attention decoder* (MAE 73.5) reduces spatial reasoning ability. Eliminating the *dual-headed density estimation* (MAE 69.2) also

Table 4

Cross-platform efficiency analysis. Performance across different hardware environments.

Platform	Device	FPS	Memory (MB)	Power (W)
Intel i5 Dual-Core	CPU	50	180	35
NVIDIA RTX 3090	GPU	92	360	65
NVIDIA Jetson Xavier NX	Edge GPU	27	280	12

Table 5

Ablation on architecture components (ShanghaiTech Part A). Lower is better.

Model Variant	MAE	RMSE	Params (M)
HybridCount-S (Ours)	52.25	85.64	0.22
w/o Scale-aware Embedding	71.80	114.60	0.21
w/o Cross-attention Fusion	68.70	109.20	0.20
w/o Graph Attention Decoder	73.50	116.80	0.17
w/o Dual-headed Density Est.	69.20	110.50	0.20
w/o Encoder Distillation Loss	60.30	117.90	0.22
w/o Graph Distillation Loss	62.20	120.20	0.22
w/o Density & Count Distill.	81.20	135.60	0.22

Table 6

Ablation on knowledge distillation components (ShanghaiTech Part A). Lower is better.

Distillation Variant	MAE	RMSE	Params (M)
HybridCount-S (Ours)	52.25	85.64	0.22
w/o Encoder Distillation	68.50	109.70	0.22
w/o Graph Distillation	68.10	108.50	0.22
w/o Depth Distillation	69.90	112.30	0.22
Only Ground Truth Supervision	74.80	116.70	0.22

decreases precision, confirming the complementary value of density and regression learning.

Overall, each architectural element contributes to accuracy and robustness, confirming the synergy among embedding, attention, and dual-headed decoding components.

5.2. Knowledge distillation analysis

We next examine the effect of each distillation loss. As shown in Table 6, removing any distillation pathway results in accuracy drops, proving their complementary nature. Encoder-level distillation enhances representation transfer (MAE 68.5 when removed), while graph-level distillation improves structured reasoning (MAE 68.1). Depth-level supervision further refines spatial consistency (MAE 69.90). Without any distillation (*Only Ground Truth*), performance falls sharply (MAE 74.80), underlining the importance of teacher-student knowledge transfer.

Hyperparameter Sensitivity. The weighting coefficients ($\alpha_{kd}, \beta_{kd}, \gamma_{kd}$) in Eq. (17) balance the contributions of encoder, graph, and depth distillation. To assess their influence, we conducted a sensitivity analysis by varying each parameter in $\{0.1, 0.3, 0.5, 0.7\}$ while ensuring $\alpha_{kd} + \beta_{kd} + \gamma_{kd} = 1$. The results in Table 7 indicate that our framework is robust to a wide range of settings. The best performance is obtained for ($\alpha_{kd} = 0.3, \beta_{kd} = 0.3, \gamma_{kd} = 0.4$), which slightly favors depth supervision. This is consistent with our design intuition: depth-aware distillation provides stronger geometric cues for density estimation, justifying the slightly larger γ_{kd} . Moreover, the gap between the best and average settings is small ($< 1.5\%$), showing that the model does not require fine-tuned coefficients.

5.3. Model size vs. performance

We analyze the accuracy-efficiency trade-off using student models of different sizes (Fig. 5). The 0.22M model achieves the best balance between accuracy and cost. Smaller models suffer large errors, while larger ones yield minor gains but higher computational demands.

6. Results and discussion

The performance of various crowd counting methods on the ShanghaiTech (SHT-A, SHT-B), UCF-QNRF, JHU, CAPRK, and PUCPR +

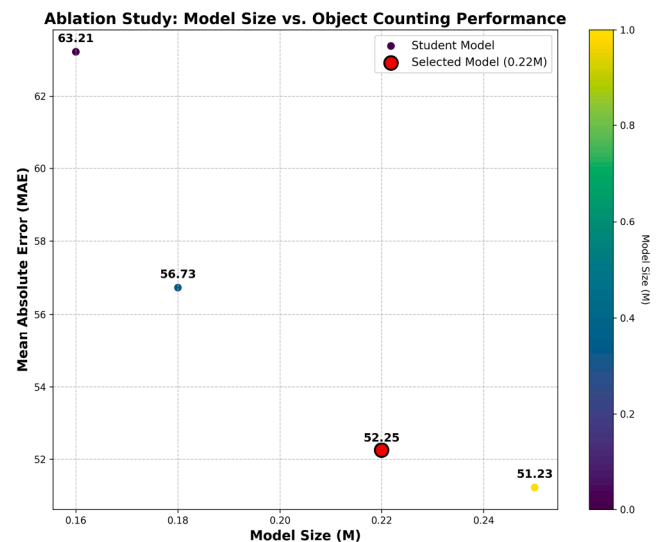


Fig. 5. Model size vs. counting performance on ShanghaiTech Part A. The 0.22M model achieves the optimal accuracy-efficiency balance.

Table 7

Hyperparameter sensitivity on validation set. Each parameter is varied while others are fixed.

	Alpha sweep ($\beta_{kd}=0.3, \gamma_{kd}=0.4$)		Beta sweep ($\alpha_{kd}=0.3, \gamma_{kd}=0.4$)		Gamma sweep ($\alpha_{kd}=0.3, \beta_{kd}=0.3$)	
Param	MAE↓	MSE↓	MAE↓	MSE↓	MAE↓	MSE↓
0.10	15.80	40.20	15.60	39.80	16.20	42.50
0.30	14.90	36.50	15.00	36.90	15.40	38.40
0.50	15.10	37.10	15.30	37.80	15.00	36.80
0.70	15.40	38.90	15.70	39.00	15.20	37.50

Table 8Performance comparison of crowd counting methods on ShanghaiTech (SHT-A, SHT-B) and UCF-QNRF datasets. Lower values are better. Best in **bold**, second-best underlined, blue indicates HybridCount-S (0.22M params).

Method	Params (M)	SHT-A		SHT-B		UCF-QNRF	
		MAE	RMSE	MAE	RMSE	MAE	RMSE
CSRNet [6]	16.26	68.20	115.00	10.60	16.00	135.40	207.40
BL [33]	21.50	61.50	103.20	7.50	12.60	87.70	158.10
UOT [34]	21.50	58.10	95.90	6.50	10.20	83.30	142.30
SUA-Fully [35]	15.85	66.90	125.60	12.30	17.90	119.50	213.30
P2PNet [36]	18.34	52.70	<u>85.10</u>	<u>6.30</u>	9.90	85.30	154.50
DKPNet [37]	30.63	55.60	91.00	6.60	10.90	81.40	147.20
MCNN [5]	<u>0.13</u>	110.20	173.20	26.40	41.30	277.00	426.00
CMTL [38]	2.47	101.30	152.40	20.00	31.10	252.00	514.00
ACSCP [39]	5.10	75.70	102.70	17.20	27.40	–	–
TDF-CNN [40]	0.13	97.50	145.10	20.70	32.80	–	–
SANet [7]	0.91	75.30	122.20	10.50	17.90	152.60	247.30
PCC-Net [41]	0.55	73.50	102.70	11.00	19.00	148.70	247.30
LCNet [42]	0.86	93.30	149.00	15.30	25.20	–	–
C-CNN [43]	0.07	88.10	141.70	14.90	22.10	–	–
LMSFFNet [44]	4.58	85.85	139.90	9.20	15.10	112.80	201.60
Distance-Aware [45]	–	59.30	98.30	6.90	10.70	98.00	170.90
FusionCount [46]	815G	62.20	101.20	6.90	11.80	–	–
Prog.Multi-Res. [47]	–	61.10	104.80	6.60	10.80	80.10	131.20
EdgeCount-T [48]	1.32	68.95	118.64	8.07	13.72	111.43	189.16
EdgeCount-S [48]	0.12	74.04	119.35	8.58	13.38	107.30	183.34
HybridCount-T (Ours)	24.00	49.95	80.64	6.07	9.34	91.20	<u>145.67</u>
HybridCount-S (Ours)	0.22	<u>52.25</u>	<u>85.64</u>	<u>7.25</u>	<u>10.43</u>	<u>93.45</u>	<u>150.31</u>

Knowledge Distillation for Crowd Counting on ShanghaiTech Dataset

Comparative Performance: Teacher vs. Student Models

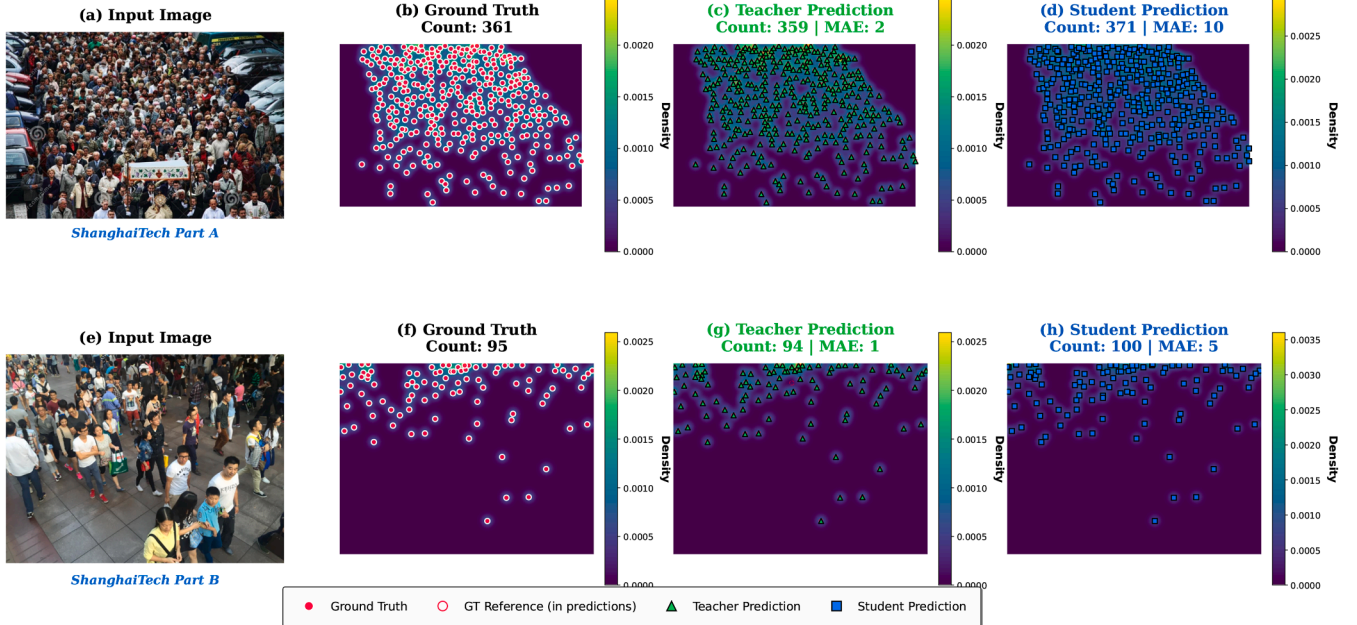


Fig. 6. Qualitative comparison of density and count predictions on ShanghaiTech Part A and Part B. The figure shows (a,e) Input Images, (b,f) Ground Truth density, (c,g) Teacher predictions, and (d,h) Student predictions. Part A (GT count = 361): Teacher predicts 359 (MAE = 2) with sharper spatial detail, Student predicts 371 (MAE = 10); Part B (GT count = 95): Teacher predicts 94 (MAE = 1), Student predicts 100 (MAE = 5), preserving similar spatial layouts. The color bar in each density panel encodes **relative crowd density magnitude per pixel region** on a normalized scale, where warmer values indicate higher person concentration and cooler values indicate lower density.

Table 9

Performance comparison on JHU dataset. FPS = frames per second. Lower MAE/RMSE is better.

Method	Params (M)	FPS	MAE	RMSE
Switch-CNN [49]	15.30	0.01	66.50	77.80
CSRNet [6]	16.26	3.92	19.80	25.60
CAN [50]	18.18	7.12	22.10	33.40
DM-Count [51]	21.50	10.04	18.40	27.00
PSGCNet [52]	27.51	6.79	24.70	31.90
CMTL [38]	2.45	2.31	56.37	65.90
LMSFFNet [44]	4.58	5.75	23.85	30.69
ACSCP [39]	5.10	1.58	48.10	60.20
EdgeCount-T [48]	1.32	15.51	24.65	29.12
EdgeCount-S [48]	0.12	18.52	20.48	28.91
HybridCount-T (Ours)	24.00	11.50	14.57	20.10
HybridCount-S (Ours)	<u>0.22</u>	17.45	<u>17.85</u>	24.30

Table 10

Performance comparison on CARPK and PUCPR+ car counting datasets. Lower is better.

Method	CARPK		PUCPR+	
	MAE	RMSE	MAE	RMSE
CSRNet [6]	11.48	13.32	8.65	10.24
BL [33]	9.58	11.38	6.54	8.13
ASPDNet [53]	7.81	10.16	–	–
PSGCNet [52]	8.15	10.46	5.24	7.36
LMSFFNet [44]	7.05	9.03	4.49	6.21
EdgeCount-T [48]	<u>6.11</u>	8.51	<u>3.56</u>	4.72
EdgeCount-S [48]	7.12	9.44	3.39	5.15
HybridCount-T (Ours)	5.89	6.63	2.34	3.89
HybridCount-S (Ours)	<u>6.15</u>	<u>7.15</u>	<u>3.43</u>	<u>4.56</u>

datasets is summarized in three tables. In Table 8, we compare state-of-the-art methods across three benchmark datasets for crowd counting. Our **HybridCount-S** model achieves remarkable results with only **0.22M parameters**, demonstrating its efficiency and effectiveness. Notably, HybridCount-S achieves competitive MAE and RMSE values, such as **52.25 MAE** on SHT-A, **7.25 MAE** on SHT-B, and **93.45 MAE** on UCF-QNRF, while maintaining minimal parameter usage. Compared to larger models like DKPNet (30.63M) and CSRNet (16.26M), our lightweight model demonstrates superior scalability without compromising accuracy. The second-best performance by methods like UOT and P2PNet highlights the trade-off between model size and performance, underscoring the significance of our approach.

In Table 9, we evaluate the performance of various methods on the JHU dataset, emphasizing both accuracy and inference speed (FPS). Our **HybridCount-S** model stands out with a balanced performance, achieving an MAE of **17.85** and RMSE of **24.30**, while running at **17.45 FPS**. This highlights its ability to handle high-density crowds efficiently in real-time applications. Compared to other lightweight models like EdgeCount-S (**0.12M**), HybridCount-S achieves better accuracy despite having slightly more parameters. Furthermore, the model outperforms larger networks such as DM-Count (21.50M) and PSGCNet (27.51M) in terms of MAE and RMSE, demonstrating that parameter efficiency does not come at the cost of performance. The results underscore the importance of designing compact yet powerful models for practical deployment in resource-constrained environments.

Finally, Table 10 evaluates the performance of various methods on car counting tasks using the CARPK and PUCPR+ datasets. Our **HybridCount-S** model achieves competitive results, with an MAE of **6.15** and RMSE of **7.15** on CARPK, and an MAE of **3.43** and RMSE of **4.56** on PUCPR+. While HybridCount-T achieves the best overall performance, HybridCount-S remains highly efficient with only **0.22M parameters**, making it suitable for edge devices or scenarios where computational resources are limited. The comparison with other lightweight models like LMSFFNet and EdgeCount-S highlights the trade-offs between accuracy and model size. These results demonstrate the versa-

Table 11

Performance gap (Student – Teacher) under challenging scenarios (UCF-QNRF).

Scenario	T-MAE	S-MAE	Δ MAE	Gap (%)
<i>Occlusion Level</i>				
Low (<30%)	48.3	49.3	+1.0	+2.1
Medium (30–60%)	56.7	61.4	+4.7	+8.3
High (>60%)	68.2	78.1	+9.9	+14.5
<i>Object Scale (pixels)</i>				
Very Small (<10)	72.5	95.8	+23.3	+32.2
Small (10–20)	58.3	67.8	+9.5	+16.3
Medium (20–50)	45.2	49.1	+3.9	+8.6
Large (>50)	38.6	39.3	+0.7	+1.8

tility of our HybridCount framework, which can be adapted to diverse object counting tasks beyond crowd counting, such as vehicle detection and monitoring.

Fig. 6 provides visual insights into the performance of our crowd counting model on the ShanghaiTech dataset, which is widely regarded as a challenging benchmark for high-density crowd analysis. The figure showcases two sample predictions: one from ShanghaiTech Part A (highly congested scenes) and another from Part B (less crowded scenes). Each subfigure includes the original image, the ground truth density map, and the predicted density map generated by our model. These visualizations demonstrate the model's ability to accurately estimate crowd density distributions, even in complex scenarios with varying levels of congestion. Notably, the predicted density maps closely align with the ground truth, highlighting the robustness and precision of our approach in capturing both fine-grained and large-scale crowd patterns. Such qualitative results complement the quantitative metrics presented in Table 8, further validating the effectiveness of our method.

6.1. Key findings and limitations

Our experiments reveal several key findings:

- Scale-aware components are critical for accurate counting, particularly in scenes with varying object scales.
- Knowledge distillation effectively transfers representational power from large to small models.
- Graph-based spatial reasoning significantly improves counting in dense regions.
- The dual-headed density estimation provides complementary signals that enhance overall counting accuracy.

While our student model achieves competitive results, we observe several limitations in complex scenes. The performance gap between the teacher and student models becomes more pronounced under extreme scale variations and heavy occlusions (e.g., in the UCF-QNRF dataset). This is primarily because the compact student network has reduced receptive field diversity, which limits its ability to simultaneously capture fine-grained small-object details and large contextual regions. Moreover, in heavily occluded regions, the graph-based feature aggregation tends to over-smooth neighboring representations, causing underestimation of densely packed objects. These observations highlight that knowledge transfer remains suboptimal when spatial cues are ambiguous or multiscale patterns dominate the scene. Future work could incorporate scale-aware feature normalization or attention-guided distillation strategies that dynamically adjust the transfer strength based on local scene complexity, thereby improving robustness in challenging visual conditions. The details of scale variation and occlusion are shown in the Table 11.

6.2. Cross-domain generalization

To assess the robustness of our framework across heterogeneous domains, we evaluate HybridCount on both crowd and vehicle count-

ing benchmarks, including ShanghaiTech Part A/B, UCF-QNRF, JHU-CROWD++, CARPK, and PUCPR+. These datasets collectively span diverse visual scenarios, ranging from high-density human crowds to structured vehicle layouts under varying viewpoints. Our model achieves consistently strong performance across all datasets without any additional fine-tuning, indicating that the learned representations capture transferable spatial and contextual cues rather than being task-specific. This highlights the model's capacity for effective cross-domain generalization in real-world counting scenarios.

7. Conclusion

This work presents a *scale-aware knowledge distillation framework* for lightweight object counting that achieves a strong balance between accuracy and efficiency. The proposed model integrates scale-aware embeddings, a graph attention decoder, and a dual-headed density-regression module to capture spatial relationships and preserve scale identity, resulting in competitive accuracy (MAE = 52.25 on ShanghaiTech Part A) with a 99.1% reduction in parameters and real-time performance of 85 FPS. While the method performs well across diverse crowd densities, it remains less effective in extremely dense or heavily occluded scenes and requires careful hyperparameter tuning. Nevertheless, the framework provides valuable insights into efficient transformer compression and offers components adaptable to other multi-scale vision tasks. Future work will focus on adaptive distillation, self-supervised pretraining, and attention mechanisms optimized for edge devices. These improvements target stronger generalization and faster, efficient deployment. Overall, this work advances practical and scalable object-counting solutions for crowd management, traffic analysis, and environmental and ecological monitoring.

CRedit authorship contribution statement

Jayanthan K S: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Domnic S:** Supervision, Resources, Project administration, Investigation, Data curation.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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